DevSync: Requirements Gathering & System Specification

Digital Egypt Pioneers Initiative (DEPI) — Project Technical Documentation

1. Stakeholder Analysis

This section identifies the primary stakeholders and their core objectives within the system:

- **Investors / Project Founders:** To deliver an innovative, high-demand technical solution for the developer ecosystem while establishing a sustainable revenue model (e.g., success fees, commission per matched collaboration, or subscription-based tiers).
- **Core Project Maintainers / Lead Developers:** Established developers who manage large open-source projects and face an influx of work due to high client trust. They require a reliable platform to discover and onboard vetted collaborating developers to delegate tasks efficiently without compromising quality.
- **Collaborating Developers:** A streamlined developer-facing platform to showcase their technical stack, discover meaningful open-source projects aligned with their expertise, and secure collaboration opportunities.

2. User Stories & Use Cases (Expanded)

User Story Set 1: Authentication & Account Management

- **As a new user (Lead or Collaborating Developer),** I want to access clean and secure registration screens so that I can create an account using my credentials or external providers (like Google).
- **As a returning user,** I want to log in securely to my dashboard so that I can manage my active projects, view matches, and continue collaborations.
- **As a platform user,** I want my session to remain secure and authenticated across devices to protect my profile and project data.

User Story Set 2: Lead / Core Developer Collaboration

- **As a lead developer managing high-demand open-source projects with strict client trust,** I want to input project properties, descriptions, and required technical skills so that the system can automatically parse and match me with qualified collaborators.
- **As a core maintainer,** I want to review a curated list of AI-suggested collaborating developers, inspect their verified GitHub activity, and select the best fit for my project.
- **As a project owner,** I want to complete the collaboration agreement smoothly and process the required platform service fee or commission securely.

User Story Set 3: Collaborating Developer Participation

- **As a developer with an active GitHub profile,** I want my real-world repository activity, commit history, and skill sets to be indexed so that I am accurately recommended for relevant collaborative projects matching my technical stack.
- **As a collaborating developer,** I want to receive real-time notifications or updates when a lead developer invites me to collaborate on a project.

3. Functional Requirements

The core features and system functionalities required for DevSync:

- **Login & Registration Screens (Authentication):** Secure user onboarding and authentication flows supporting multi-platform sign-in (e.g., email/password and Google authentication) to manage user identity and profile data.
- **Project Properties & Details Management:** Comprehensive interface to define project scope, descriptions, deliverables, and requirements.
- **Required Technical Stack / Skills Specification:** Dedicated input parameters to specify exact programming languages, frameworks, and developer tooling needed.
- **GitHub Link Integration:** Repository and profile link ingestion used as the source of truth for developer activity parsing.
- **AI Agent Matching Engine:** An intelligent core service that ingests project parameters and required skills, queries GitHub data, and computes a high-precision match score between developers and projects.
- **Suggested Collaborators Dashboard:** A curated, ranked list of recommended collaborating developers generated by the AI matching algorithm.
- **Monetization & Success Fee Payout:** A secure in-app payment gateway or profit-sharing mechanism triggered upon final collaborator selection and collaboration confirmation.

4. Non-Functional Requirements

Quality attributes, security protocols, and performance criteria governing the system:

- **Performance:** Streamlined and frictionless data input workflows, along with low-latency search and real-time AI matching response times.
- **Security:** End-to-end encryption, secure token management, and strict data privacy compliance for user credentials, proprietary project details, and matching query results.
- **Usability:** Clean, intuitive UI/UX designed with a low learning curve for rapid project and skill configuration.
- **Reliability & Trust Criteria:** Direct integration with verified GitHub profiles showcasing proven track records, contribution graphs, technical expertise, and historical open-source project participation.